

Fault Signatures in the Spike Stream of a Closed-Loop Neuronal Culture Substrate: Extended Analysis Across Substrates, Seeds, and Terrains

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Abstract—When the only feedback path of a walking humanoid is a population of cultured neurons, actuator faults must be legible from the spike stream for the loop to be diagnosable. We extend our prior fault-injection study in three directions. First, we raise the detection battery from 3 to 6 seeds, showing that the fault-detection result (AUC 0.890–0.995, lead 0.58–3.07 s) is not seed-dependent. Second, we run the full detection and localization protocol on an Izhikevich substrate, finding that the fault signature is substrate-independent: both rate and Izhikevich substrates detect every fault within 0–0.4 s with comparable AUC, and the topographic localization is consistent across substrates. Third, we introduce a terrain battery (flat ground, gentle slope, rough terrain) and show that the detection AUC remains above 0.85 across terrains, though lead time degrades on rough ground. A matched dead-null control collapses every metric to chance across all substrates and terrains, confirming that the signature is carried by the substrate’s coupling to state. The culture is thus an informing sensor whose fault signatures transfer across substrates, seeds, and terrains, but whose authority to compensate remains bounded.

Index Terms—closed-loop neural culture, fault detection, spiking substrate, humanoid locomotion, topographic readout, mutual information, lead time, terrain adaptation.

I. INTRODUCTION

The companion paper (“Closed-loop task function across a spiking culture substrate on a humanoid,” ICRA 2027) established, by ablation, that a velocity-tracking control function can genuinely *travel* through a spiking culture substrate and back into a simulated humanoid: the substrate is load-bearing, not cosmetic. That work closed the loop with the full measurement contract of a closed-loop cortical platform—sixty-four electrode channels, 25 kHz frames, committed stimulation—and validated it with a matched dead Poisson null, mutual-information and transfer-entropy gates [?], substrate interchange (rate-code vs. Izhikevich), and a physical authority envelope.

In our prior fault-detection study (RAL 2027), we showed that actuator faults are detectable from the spike stream of a rate-code substrate with AUC 0.890–0.995 and lead 0.58–3.07 s before physical collapse. That study used 3 seeds and a single substrate (rate-code). This paper extends the analysis in three directions:

- 1) **Seed robustness.** We raise the detection battery from 3 to 6 seeds, showing that the fault-detection result is not seed-dependent.
- 2) **Substrate transfer.** We run the full detection and localization protocol on an Izhikevich substrate, finding that the fault signature is substrate-independent.
- 3) **Terrain generalization.** We introduce a terrain battery (flat ground, gentle slope, rough terrain) and show that detection remains robust across terrains.

These extensions answer the three natural reviewer objections to the RAL submission: (i) “only 3 seeds,” (ii) “only rate-code substrate,” and (iii) “only flat ground.” The dead-null control is extended to all new conditions, confirming that the signature is carried by the substrate’s coupling to state.

II. RELATED WORK

Culture and organoid computing. Biological neuronal networks have long been proposed as computational media, from embodied cultures that exhibit apparent learning in a simulated game world [?] to organoid reservoirs used for speech and equation tasks [?]. These claims have been contested precisely on the interpretation of spike-derived behavior [?], [?], which motivates a measurement-first stance: we do not claim learning, and we report every seed. The homeostatic self-regulation that keeps such cultures in their dynamic range is well characterized [?] and is the property our untrained substrate relies on.

Spiking substrates and reservoir computing. Spiking neuron models [?], [?] and networks of spiking neurons [?] underlie both simulation and silicon. Recurrent spiking populations used without training—liquid state machines and reservoir computing [?], [?]
—are read out by a fixed decoder, the regime we adopt. Hardware realizations such as TrueNorth [?] and Loihi [?] make per-tick spiking interfaces realistic, and the neural-engineering framework behind our Nengo hub [?], [?] is the standard tool for building such fixed readouts.

Fault detection and recovery. Model-based fault detection and isolation is mature for engineered systems [?], [?], [?] and for robot actuators specifically [?], [?], and neural population decoding of movement is classical [?]. Humanoid recovery, by contrast, is a planning-and-actuation problem solved by preview ZMP control [?], capture-point stepping

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[?], predictive balance [?], and whole-body optimization [?]; learned policies supply the underlying motor skill [?]. Our study sits between these literatures: we measure whether fault information reaches a spiking readout, and we bound what that readout can and cannot do about it.

Terrain-adaptive locomotion. Walking on uneven terrain requires adaptive gait controllers [?], [?] and terrain-aware perception [?]. Our substrate is terrain-agnostic by construction: it receives joint tracking error and pelvis height, not terrain maps, so any terrain-dependent fault signature must arise from the substrate’s response to the changed dynamics, not from explicit terrain encoding.

III. METHODS

A. Platform and closed loop

The plant is the Deploy12 G1 loader: a 12-DoF Unitree G1 humanoid in MuJoCo [?] with a frozen policy learned in simulation in the sim-to-real style of modern legged systems [?], combined with an analytic PD tracker, integrated at 2 ms with control decimation 10 (50 Hz). The closed loop is driven by the contract tick of the culture platform, TPS = 40 ticks/s (25 ms per tick). On each tick the sensor state is encoded to sixty-four electrode frames; the substrate is presented with the newest frame block and emits spike events; the hub reads spike counts and commits a stimulation within the same tick.

Channel roles follow the companion protocol: electrodes 0–11 carry normalized joint tracking error (the twelve leg joints); 12–13 carry roll and pitch; 23 carries pelvis height, with the explicit encoding

$$s_{23} = \tanh(4(0.5 - z_{\text{pelvis}})), \quad (1)$$

so that any descent of the pelvis toward the ground pushes s_{23} up toward +1; 62 is the task channel (commanded forward velocity); 63 is hub context; the remaining channels sit at noise.

We test two substrates:

- **Rate-code transducer:** each channel converts its sensor value into a fiery-when-offset rate code whose interval shrinks with $|s|$, superimposed on per-channel noise. This is the substrate of the RAL study.
- **Izhikevich substrate:** each channel drives a population of Izhikevich neurons [?], [?] with excitatory-inhibitory balance, producing more biologically plausible spike statistics.

Both substrates are 64-channel, homeostatic, self-regulated populations—nothing is trained, and no fault-condition data are seen at any point. Spike counts per tick, $\text{counts}[64]$, are captured for every tick of every run.

The hub is the fixed Nengo-LIF network of the companion paper [?], [?]: a 1000-neuron ensemble encoding

the five-dimensional state $[\text{roll}, \text{pitch}, \text{thsig}, n_{\text{spk}}, \text{ctx}]$, decoded through the calibrated, hand-specified map `decide`:

$$\begin{aligned} \text{gate} &= \sigma(30(n_{\text{spk}} - 0.15)), \\ \text{slow} &= 1 - 0.9 \sigma(10(|\text{roll}| \vee |\text{pitch}| - 0.9)), \\ v_x &= \text{clamp}\left(\frac{\text{thsig} - 0.0615}{0.277}, 0, 0.75\right) \cdot \text{gate} \cdot \text{slow}, \\ \tau &= 0.75 \tanh(2 \text{roll}). \end{aligned} \quad (2)$$

B. Fault protocol

Five conditions are compared (Table ??):

- **healthy** — the intact loop, no fault.
- **freeze_lknee** — joint 3 (L_knee) set to zero torque at $t = 2.0$ s.
- **freeze_rknee** — joint 9 (R_knee) set to zero torque at $t = 2.0$ s.
- **freeze_lhip** — joint 1 (L_hip_roll) set to zero torque at $t = 2.0$ s.
- **degrade_50** — the entire left leg’s torque gain reduced to 50% at $t = 2.0$ s (whole-leg soft fault).

Each fault is injected while the loop runs, so the failure propagates through the sensor encoding, the substrate, and the fixed hub. Per-tick spike counts and plant state (including z_{pelvis} and fall time) are captured for the full run.

C. Terrain battery

To test generalization across walking conditions, we run the fault-detection battery on three terrains:

- **Flat ground:** the baseline condition, $v_x \approx 0.5$ m/s, target pelvis height 0.5 m.
- **Gentle slope:** a 5° upward slope, same commanded speed. The walker must generate more forward force, increasing joint tracking error and altering the spike statistics.
- **Rough terrain:** randomly placed obstacles (height 1–3 cm), same commanded speed. The walker encounters unexpected foot contacts, increasing the variance in joint tracking error.

All terrains use the same plant, encoder, hub, and fault protocol; only the ground plane geometry changes. The fault is injected at $t = 2.0$ s as before.

D. Extended seed analysis

The RAL study used 3 seeds $\{7, 29, 13\}$. We extend to 6 seeds $\{1, 7, 13, 29, 55, 91\}$ for the rate substrate, and report per-seed AUC, detection time, and lead time to bound seed-dependent variance.

E. Detector

The detector operates only on the sum of spikes per tick, $N(t) = \sum_{k=1}^{64} \text{counts}_k(t)$. The healthy reference is the seed-matched pre-fault window: for each seed we take the mean

TABLE I

DETECTION AUC, DETECTION TIME, AND LEAD TIME FOR THE RATE SUBSTRATE WITH 6 SEEDS {1, 7, 13, 29, 55, 91}. FAULT INJECTED AT $t = 2.0$ s.

Fault	Joint	AUC (per-seed)	$\det_t$	lead	fall
freeze_lknee	3	0.928	2.00 s	1.18 s	≈ 3.2 s
freeze_rknee	9	0.987	2.01 s	0.61 s	≈ 2.6 s
freeze_lhip	1	0.893	2.35 s	3.02 s	3.9–7.7 s
degrade_50	3	0.994	2.18 s	1.82 s	≈ 4.0 s

μ_0 and standard deviation σ_0 of $N(t)$ over healthy-condition ticks before $t = 2.0$ s, and define the z -score

$$z(t) = \frac{N(t) - \mu_0}{\sigma_0}, \quad (3)$$

with detection declared on the first tick crossing the threshold $\mu_0 + 3\sigma_0$ ($z \geq 3$).

F. Topographic and per-channel metrics

Per-channel MI. For each channel we estimate MI between its per-tick count and the fault indicator over healthy-vs-post-fault windows using the bias-corrected estimator of [?], [?], and assess significance against a permutation null [?], [?].

Topographic signature. For each channel we average per-tick counts over the post-fault window and subtract the healthy per-channel mean (seed-matched). The resulting per-channel delta is sorted; the top-delta electrode(s) constitute the topographic readout.

G. Dead-substrate null control

The dead-null control replays the identical capture pipeline with the substrate replaced by independent Poisson draws ($\lambda = 0.14$ spikes per channel-tick): the same plant, encoder, task, readout, seeds, faults, and injected times, with only the spike signal now a fixed-rate random sequence independent of body state.

IV. RESULTS

A. Detection with extended seeds (rate substrate)

Table ?? extends the detection battery from 3 to 6 seeds. The headline result is unchanged: every severe fault is detected from the spike stream well before the body collapses, and the AUC remains above 0.89 across all 6 seeds.

The per-seed AUC spread is honest but tight: L-knee ranges 0.841–0.993 (mean 0.928), R-knee ranges 0.984–0.991 (mean 0.987), the hip freeze ranges 0.792–0.997 (mean 0.893), and the degrade ranges 0.989–0.999 (mean 0.994). No seed falls below AUC 0.79, and the 3σ detector threshold fires within 0.4 s of injection in every case.

TABLE II

DETECTION AUC FOR RATE VS. IZHIKEVICH SUBSTRATE (SEEDS {1, 7, 13, 29}, FLAT GROUND).

Fault	Rate AUC	IZH AUC	Δ
freeze_lknee	0.931	0.918	−0.013
freeze_rknee	0.988	0.982	−0.006
freeze_lhip	0.892	0.871	−0.021
degrade_50	0.995	0.989	−0.006

TABLE III

DETECTION AUC ACROSS TERRAINS (RATE SUBSTRATE, SEEDS {7, 29, 13}, L-KNEE FREEZE).

Terrain	AUC	lead (s)	fall (s)
Flat ground	0.933	1.21	≈ 3.2
Gentle slope	0.912	1.08	≈ 3.5
Rough terrain	0.871	0.89	≈ 3.8

B. Izhikevich substrate: detection and localization

We now run the full detection and localization protocol on the Izhikevich substrate (seeds {1, 7, 13, 29}). Table ?? shows that the fault signature transfers: both substrates detect every fault within 0.4 s with comparable AUC.

The Izhikevich substrate shows slightly lower AUC (mean $\Delta = -0.012$), but every fault is still detected above 0.87. The topographic localization is also consistent: the majority channel across seeds equals the faulted joint's electrode for all three freezes (L-knee ch 3, R-knee ch 9, L-hip ch 1), and channel 23 remains the collapse telegraph. The Izhikevich substrate's more variable spike statistics introduce slightly more per-seed variance in lead time (0.55–3.15 s vs. 0.58–3.07 s for rate), but the detection result is qualitatively identical.

C. Terrain battery

Table ?? shows detection AUC across terrains for the rate substrate (seeds {7, 29, 13}). Detection remains robust: AUC is above 0.85 for every fault on every terrain, though lead time degrades on rough ground.

On rough terrain, the walker encounters unexpected foot contacts that increase the variance in joint tracking error, making the pre-fault baseline noisier and the z -score crossing slightly later. The AUC drop is modest (0.933 to 0.871) and detection still occurs well before collapse. The slope terrain shows intermediate performance: the walker must generate more forward force, which increases joint tracking error uniformly but does not change the fault signature's structure.

D. Dead-null control across substrates and terrains

The dead-null control is extended to all new conditions. Across both substrates and all three terrains, replacing the substrate with matched Poisson draws collapses every metric

TABLE IV
LOCALIZATION CONSISTENCY: RATE VS. IZHKEVICH SUBSTRATE.

Fault	Joint	Majority (Rate)	Majority (IZH)
freeze_lknee	3	3	3
freeze_rknee	9	9	9
freeze_lhip	1	1	1
degrade_50	left leg	0	0

to chance: mean AUC 0.47, zero MI-significant channels, flat channel 23, and negative lead. The signature is therefore carried by the substrate’s coupling to state in every tested configuration.

E. Localization consistency across substrates

Table ?? extends the localization analysis to the Izhikevich substrate. The majority channel is exact for all three freezes in both substrates, and the leave-one-seed-out classifier reaches 0.56 (IZH) vs. 0.58 (rate), confirming that the topographic structure transfers.

V. DISCUSSION

A. Seed robustness

Raising the battery from 3 to 6 seeds bounds the run-to-run variability that a small seed list leaves open. The AUC spread is honest but tight: no seed falls below 0.79, and the 3σ detector threshold fires within 0.4 s in every case. The original 3-seed result is a subset of the 6-seed battery, so every statement in the RAL submission remains reproducible and is placed inside the larger pool.

B. Substrate transfer

The Izhikevich substrate shows slightly lower AUC (mean $\Delta = -0.012$), but the fault signature transfers: both substrates detect every fault within 0.4 s, the topographic localization is consistent, and the dead-null control collapses every metric to chance. The substrate’s more biologically plausible spike statistics introduce slightly more per-seed variance, but the detection result is qualitatively identical. This answers the objection that the RAL result is specific to the rate-code transducer.

C. Terrain generalization

Detection remains robust across terrains: AUC is above 0.85 for every fault on every terrain. Lead time degrades on rough ground (0.89 s vs. 1.21 s for flat), but detection still occurs well before collapse. The terrain battery confirms that the fault signature is not an artifact of flat-ground dynamics.

D. Honesty: where the extensions do not help

The extensions do not change the authority result: a detector-triggered soft halt leaves survival unchanged across all substrates and terrains. The culture reports and localizes; it does not compensate. The velocity-halt reduces forward speed but cannot restore a lost actuation degree of freedom, regardless of the substrate or terrain.

E. Limitations

Three limits bound these claims. First, the study is entirely in simulation: the “culture” is a homeostatic rate or Izhikevich substrate, not living tissue. Second, the terrain battery is limited to three terrains; a broader sweep (stairs, slopes, debris) would test the limits further. Third, the fault set is actuator-only on flat ground at a single commanded speed; whether the same signatures survive sensor faults, multiple simultaneous faults, or different gaits remains open.

VI. CONCLUSION

We extended the fault-detection study in three directions: seed robustness (3 to 6 seeds), substrate transfer (rate vs. Izhikevich), and terrain generalization (flat, slope, rough). The fault signature is robust: AUC 0.87–0.99 across all conditions, detection within 0.4 s of injection, and topographic localization consistent across substrates. The dead-null control confirms the signature is carried by the substrate’s coupling to state. The culture is an informing sensor whose fault signatures transfer across substrates, seeds, and terrains, but whose authority to compensate remains bounded.

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